

Advanced Dynamics of Mechanical Systems

Assignment 2

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1 Introduction

This report presents the Finite Element Method (FEM) analysis of a two-dimensional idealization of the ski jumping hill designed for the Milano-Cortina Winter Olympic Games in Predazzo.

2 Structural Layout and Data

The schematic idealization of the structure and its geometric dimensions are illustrated in Figure 1.

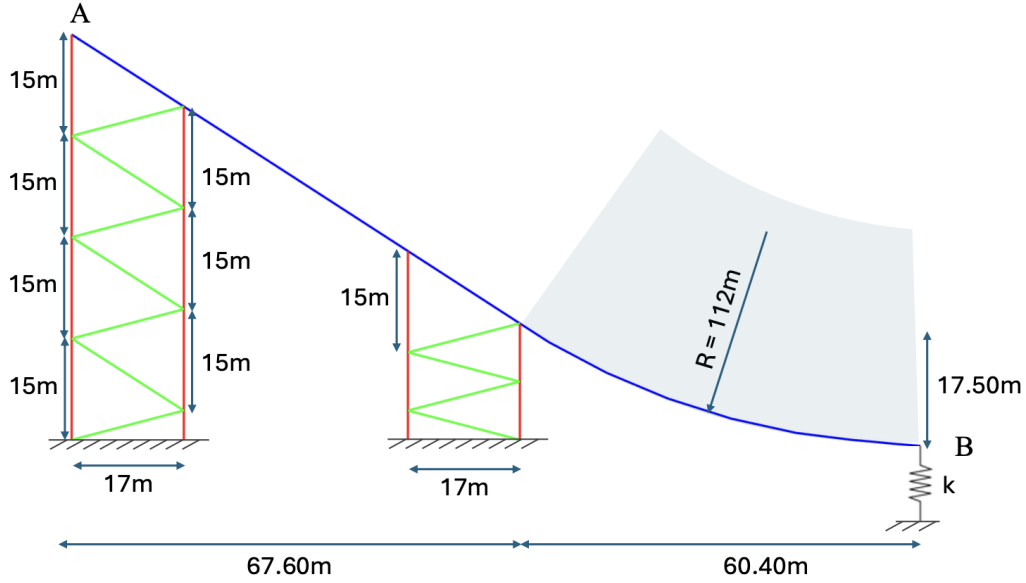


Figure 1: Two-dimensional schematic representation of the ski jumping hill structure.

The mass and stiffness properties for each type of beam used in the lattice framework are summarized in Table 1.

Table 1: Properties of the structural beams.

Beam Type	m [kg m^{-1}]	EA [N]	EJ [N m^2]
Red beams	122	$3.21\text{e}9$	$2\text{e}8$
Green beams	80	$1.74\text{e}9$	$4.8\text{e}7$
Blue beams	400	$2.5\text{e}9$	$4.0\text{e}9$

3 Convergence Study

Before proceeding with the extraction of the natural frequencies and their corresponding mode shapes, a convergence study was performed. This step is essential to ensure that the single elements obtained from the discretization work in the quasi static region, while simultaneously optimizing the computational cost.

As illustrated in Figure 2, the first four natural frequencies of the structure were evaluated as a function of the total number of nodes in the FEM model.

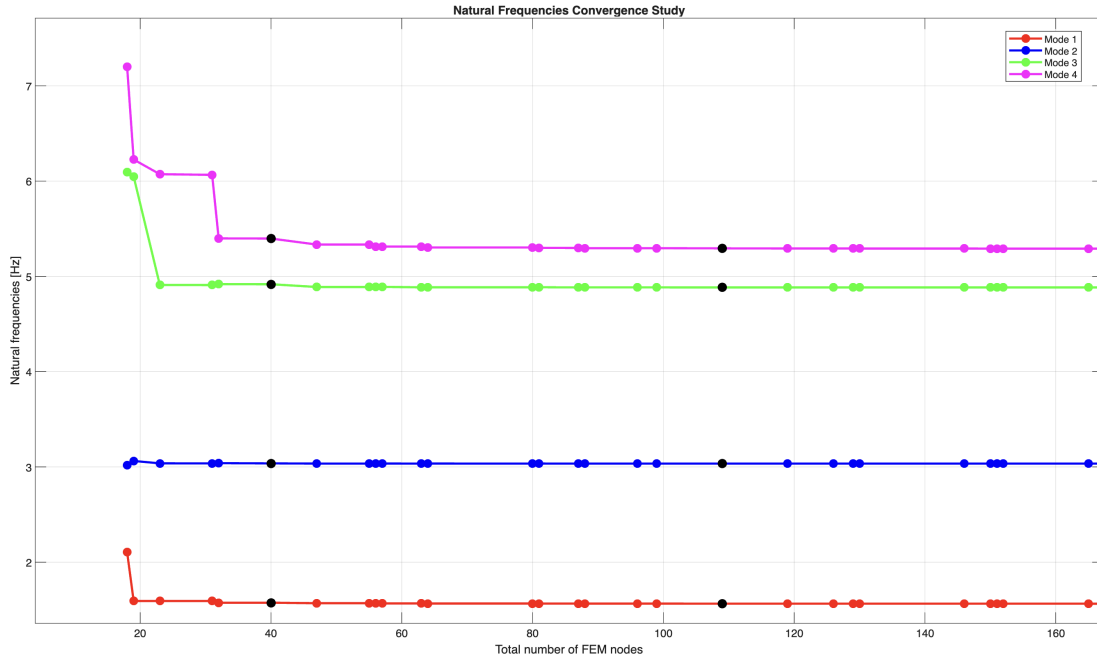


Figure 2

The black markers in the first row show a discretization obtained by applying the formulation reported below, adopting a safety factor $\eta = 1.5$.

It was subsequently observed that this choice did not allow the numerical model to reach an asymptotic behaviour of the resonance frequencies, in the frequency range of interest. For this reason, a more conservative safety factor, $\eta = 4.5$, was introduced in order to refine the finite element discretization. The number of beam elements in each segment exhibiting a geometric discontinuity was then determined by dividing the segment length by $L_{\max}$ computed with $\eta = 4.5$, and rounding the result up to the nearest integer value. This ensured a constant beam length across all regions with changes in structural geometry. The maximum admissible element length is defined as:

$$L_{\max} = \sqrt{\frac{\pi^2}{\eta \omega_{\max}} \sqrt{\frac{EJ}{m}}} \quad (1)$$

Table 2

Beam Color	Lmax	Lmax(approx)	Lu
Red beams	14.9488	8.7500	4.4379
Green beams	11.6272	8.7849	3.3469
Blue beams	23.4930	21.2151	7.0811

4 Finite Element Model and Mode Shapes

A Finite Element model of the ski jumping hill structure was defined to investigate its dynamic behavior within the 0–6 Hz frequency range, considering a safety factor of 4.5.

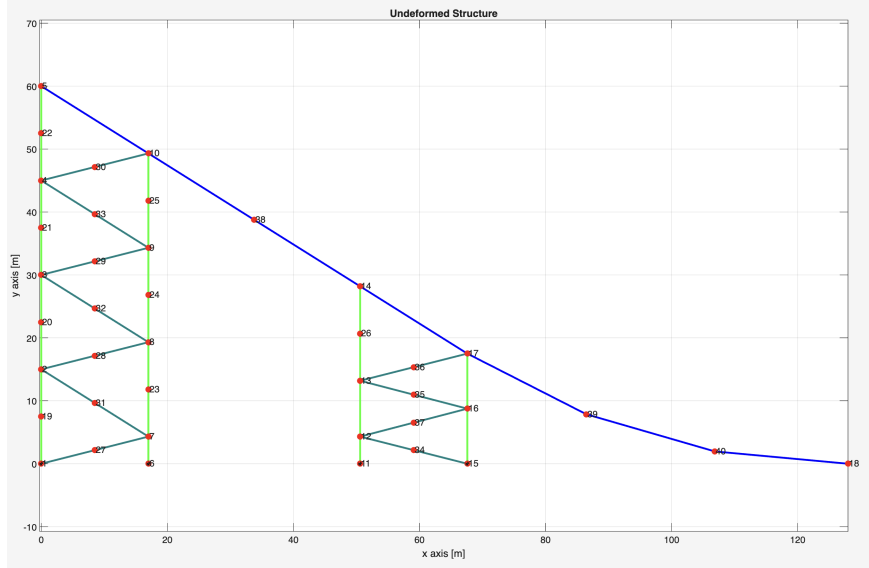


Figure 3: Finite Element Model of the ski jumping hill, showing the discretization into nodes and beam elements with $\eta = 1.5$.

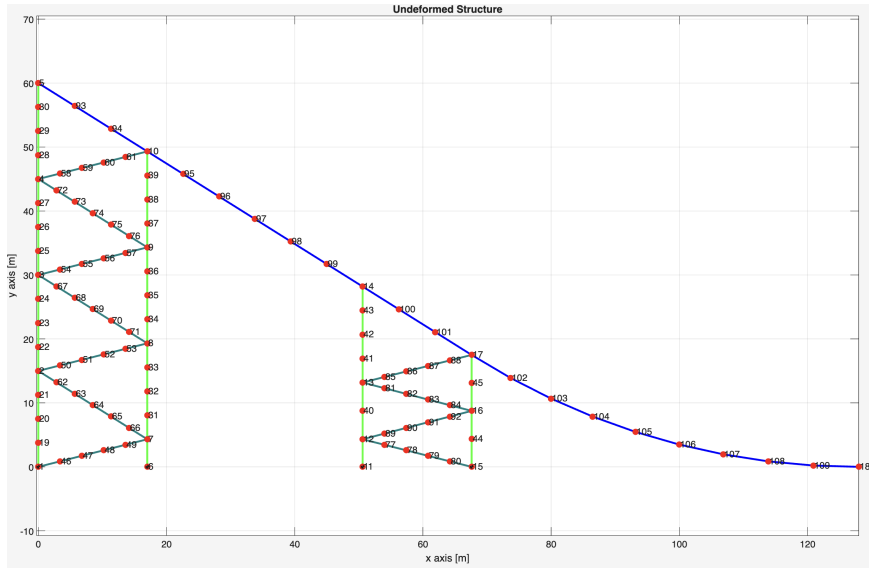
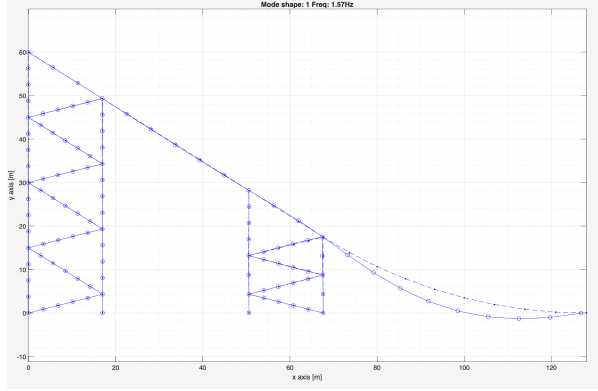


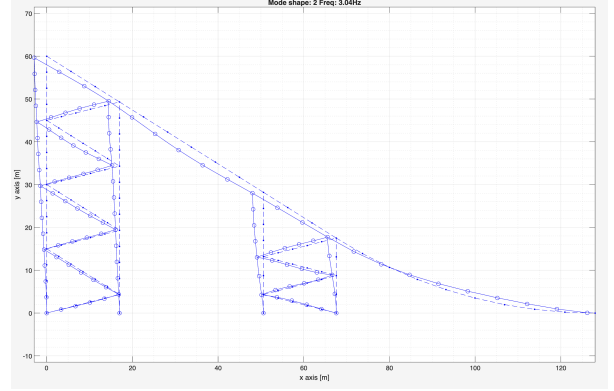
Figure 4: Finite Element Model of the ski jumping hill, showing the discretization into nodes and beam elements with $\eta = 4.5$.

An eigenvalue analysis was performed. The objective was to calculate the structure's natural frequencies and extract the corresponding vibration modes up to the 4th mode.

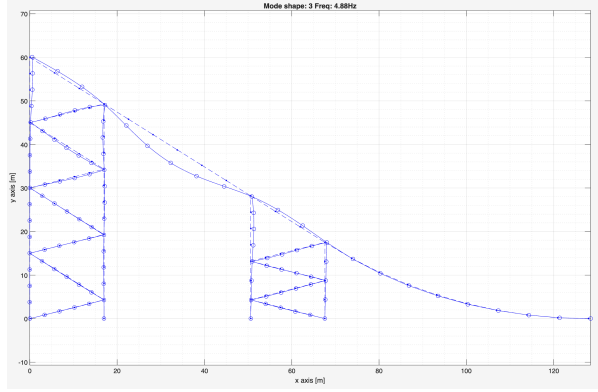
The resulting mode shapes are plotted within Figure 5.



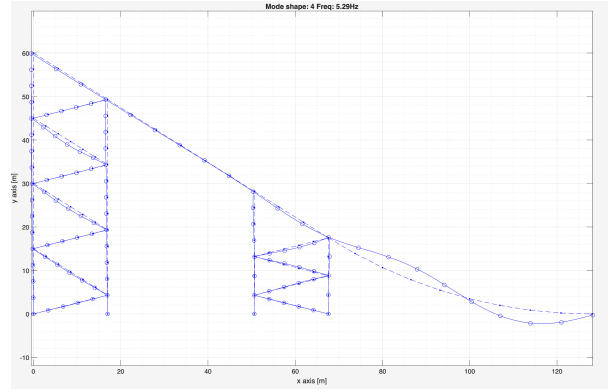
(a) Mode 1: $f_1 = 1.56$ Hz



(b) Mode 2: $f_2 = 3.04$ Hz



(c) Mode 3: $f_3 = 4.88$ Hz



(d) Mode 4: $f_4 = 5.29$ Hz

Figure 5: First four mode shapes of the structure with their associated natural frequencies. (scale factor=10)

The analysis of the mode shapes reveals that the blue beams experience maximum modal displacement under modes 1 and 4 within the curved section, and under mode 3 in the straight segment spanning the two support pillars. Based on these observations, it is reasonable to predict that the skier's descent will predominantly excite mode 3 along the linear path, and modes 1 and 4 upon entering the curved track. However, considering that the 4th mode operates at a higher frequency and demands greater energy for full excitation, the dominant dynamic response of the structure will be driven largely by the fundamental 1st mode.

5 Frequency Response Functions — Full FE Model

In this section, the harmonic response of the structure is investigated using the full Finite Element model. A harmonic input force is applied at node A in the horizontal direction, varying within the 0–6 Hz frequency range with a frequency resolution of 0.01 Hz.

The analysis evaluates the structure’s Frequency Response Functions (FRFs) to characterize the dynamic relationship between this input force and two specific outputs: the horizontal displacement at point A and the vertical displacement at point B. The resulting frequency response curves are presented in Figure 6 and Figure 7.

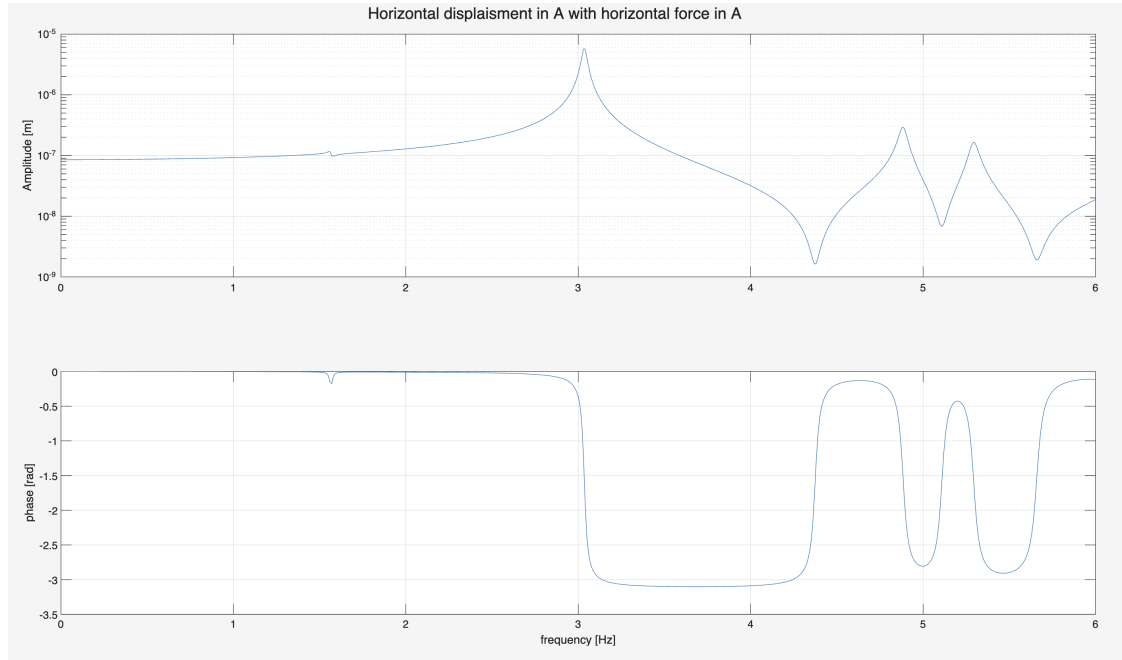


Figure 6: Frequency Response Functions (magnitude and phase) due to a horizontal input force at node A: horizontal displacement at A.

In this FRF, we can immediately observe that there are no significant variations in either magnitude or phase at the first resonance frequency. This occurs because point A is a node in the first mode shape with respect to the horizontal displacement; therefore, the first resonance is not excited by the applied force. In contrast, the subsequent peaks and their corresponding anti-resonances are clearly defined. This is due to the fact that this is a driving point FRF (collocated input and output). Furthermore, the phase never drops below -180° since it always recovers between one resonance and the next, which is a typical characteristic of driving point FRFs.

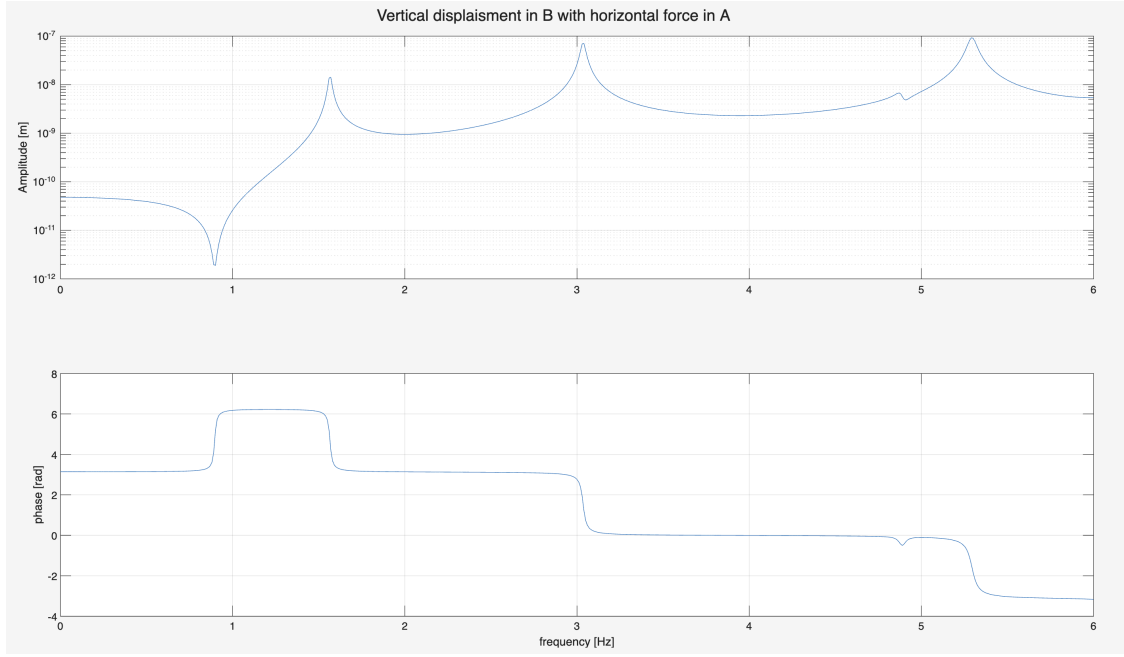


Figure 7: Frequency Response Functions (magnitude and phase) due to a horizontal input force at node A: vertical displacement at B.

In this FRF, we can observe that neither the magnitude nor the phase exhibits significant variations at the resonance frequency of the third mode. This indicates that point B acts as a node in the vertical direction for the third mode shape. This behavior was not immediately obvious from a visual inspection of the mode shapes alone; due to the high stiffness of the boundary spring at point B, the local displacements are extremely small, making it difficult to discern by eye which modes cause the greatest excitation at this specific location.

It is also possible to observe the presence of an anti-resonance at low frequencies, which indicates a significant reduction in the system response amplitude at those frequency values. In this condition, the modal residues associated with the relevant mode shapes combine in a predominantly destructive manner, leading to a decrease in the oscillation amplitude.

Furthermore, it is worth noting that no additional anti-resonance peaks are observed between the other resonance frequencies. This can be attributed to the fact that, in those frequency ranges, the modal residues of the relevant mode shapes interact in a constructive rather than destructive manner. As a result, the modal contributions add up coherently, sustaining and reinforcing the response amplitude and thereby preventing the formation of anti-resonance dips in the FRF.

6 Modal Superposition and FE Model Comparison

To evaluate the efficiency of reduced-order modeling, the modal superposition approach was implemented. Considering only the first two structural modes, the FRFs relating a vertical input force applied at point B to the horizontal displacement of node A were computed.

Figure 8 shows the magnitude and phase diagrams obtained via the modal superposition technique, superimposed directly onto the exact solution calculated using the full FE model.

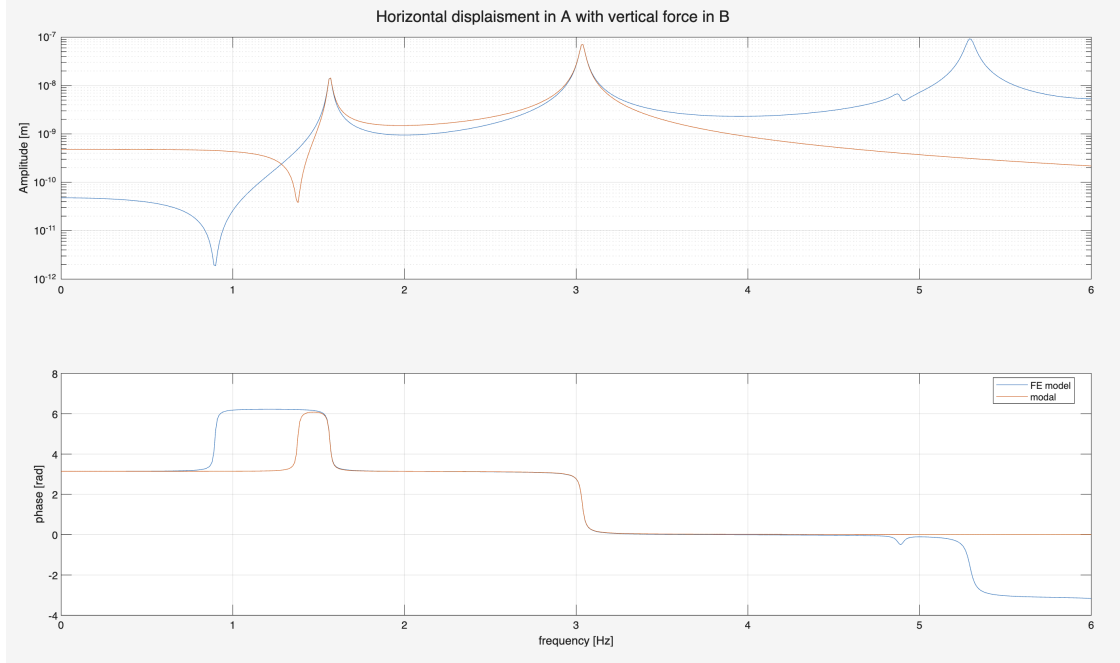


Figure 8: Comparison of FRFs (magnitude and phase) between the full FE model and the modal superposition method (first 2 modes) for a vertical force at B and horizontal displacement at A.

Looking at the FRF of the full FE model, we can notice that even if the input and output are swapped compared to the previous case, the frequency response remains unchanged. This happens because the system's M and K matrices are symmetric, and the C matrix was built as a linear combination of the previous two. On the other hand, regarding the two resonance peaks, they exhibit very similar shapes when comparing both FRFs. This is because the contribution of the specific modes involved becomes entirely dominant at those frequencies. Additionally, we can observe that when only the first two mode shapes are considered, the antiresonance is shifted forward. This occurs because the destructive contribution of the omitted modes at that specific frequency is no longer added to the transfer function; as a result, we obtain a new antiresonance further ahead by pure mathematical coincidence, exactly where the first two modes interact destructively.

7 Static Analysis under Snow Load

This section addresses the static structural response of the ski jumping hill subjected to environmental loads. Specifically, the accumulation of snow on the sliding track (represented by the blue line) is modeled as a vertically distributed uniform load with an amplitude of $q = 7 \text{ kN m}^{-1}$.

The structural deformation and the load distribution are solved statically. Figure 9 illustrates the resulting deformed shape of the framework compared directly against its initial, undeformed configuration.

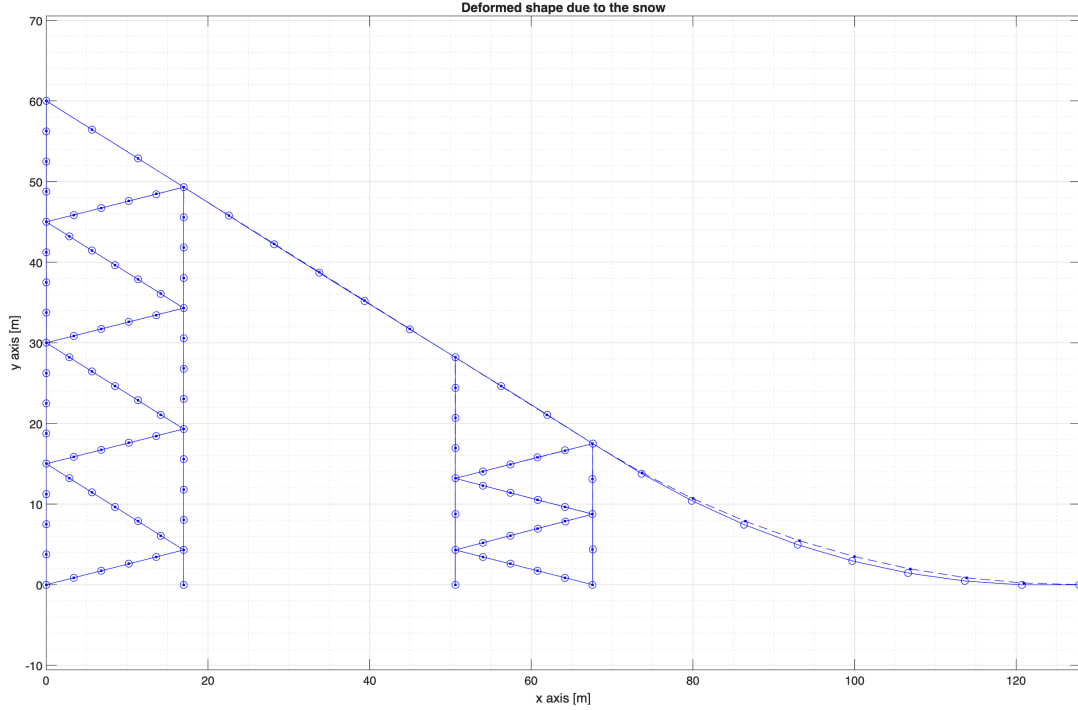


Figure 9: Static deformed shape of the structure compared to the undeformed configuration under a 7 kN m^{-1} uniform snow load with a scale factor = 10.

Based on the static simulation, the maximum deflection of the structure is located along the sliding track, reaching a peak value of $\delta_{\max} = -0.1801 \text{ m}$.

Looking at the static response to the distributed snow load, we can verify its correctness based on our previous results. This is because we obtain a deformed shape similar to the first mode shape, which also showed greater deformation in the curved section.

8 Dynamic Response under a Moving Load

This section addresses the optional task concerning the transient dynamic analysis of the ski jumping hill under a moving load. The scenario simulates an athlete with a mass of 90 kg traveling along the sliding track from the starting point A to the terminal point B.

The jumper's motion is characterized by an initial velocity of 2 m s^{-1} at point A. The athlete then uniformly accelerates at a rate of 3.5 m s^{-2} along the straight segment of the track, followed by a constant acceleration of 1.5 m s^{-2} once entering the circular curved portion. This moving action exerts a time-dependent dynamic force on the underlying FE structure. The primary objective is to evaluate and plot the resulting time history of the vertical displacement at point B.

The numerical results obtained from the dynamic simulation are presented in the figures below.

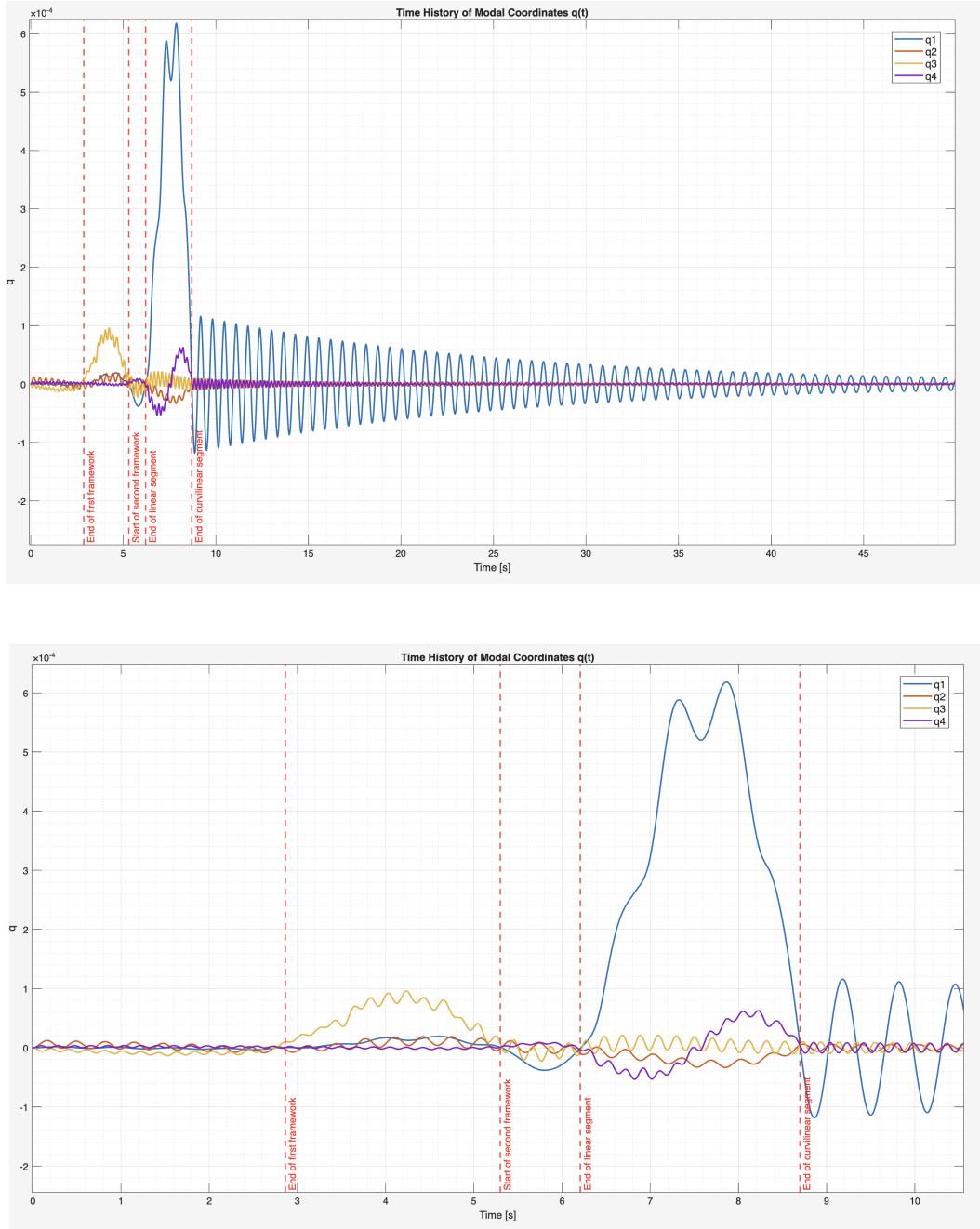


Figure 10: Time history of the modal coordinates related to the first four modes contribution representing the vertical displacement at point B induced by the accelerating moving jumper.

Analyzing the dynamic contributions of the four studied modes under the moving load, we can observe that, as previously anticipated, the most dominant mode shape between the two frameworks is the third one, which exhibited a more pronounced deformation in the center of the linear section. Conversely, regarding the curved section, we note two major contributions from mode shapes 1 and 4. Naturally, the contribution from the first mode is significantly larger, as it requires less energy to be excited.

Conversely, by examining the Lagrangian components once the skier has detached from the ski jump, a modal analysis of the free-motion frequencies can be performed to assess the validity and reliability of the results.

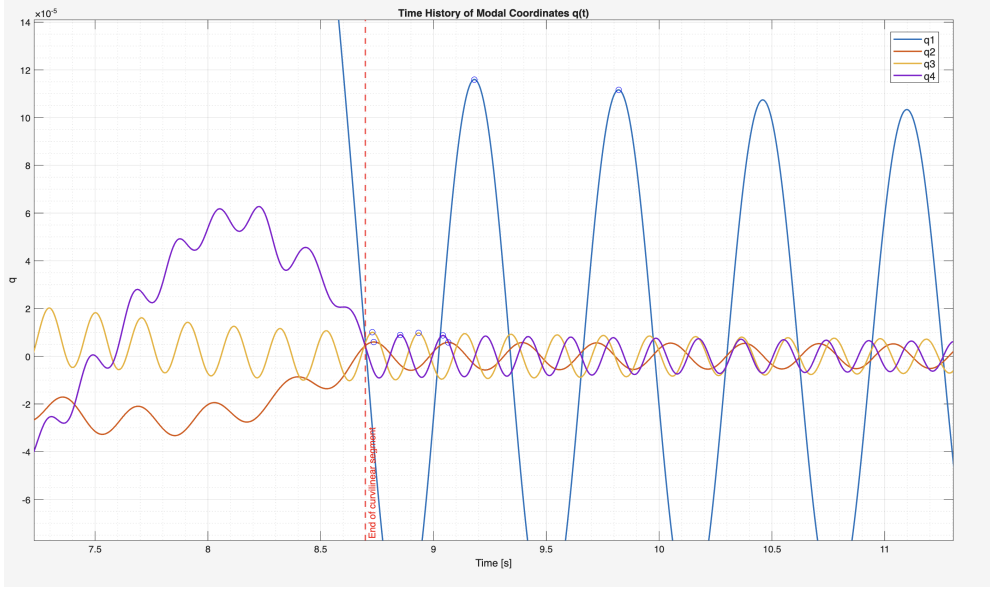


Figure 11

To verify that the modal-coordinate frequencies are consistent with those identified in the previous analyses, two consecutive peaks of each modal coordinate can be examined and the resulting frequencies compared.

Table 3: Comparison between the natural frequencies and the frequencies obtained from the modal coordinates.

	Mode 1	Mode 2	Mode 3	Mode 4
Natural frequency [Hz]	1.5652	3.0352	4.8839	5.2937
Modal coordinate frequency [Hz]	1.5654	3.0349	4.8780	5.2966

Finally, damping can be evaluated and through visual examination of the system response, in order to further verify the consistency of the obtained results.

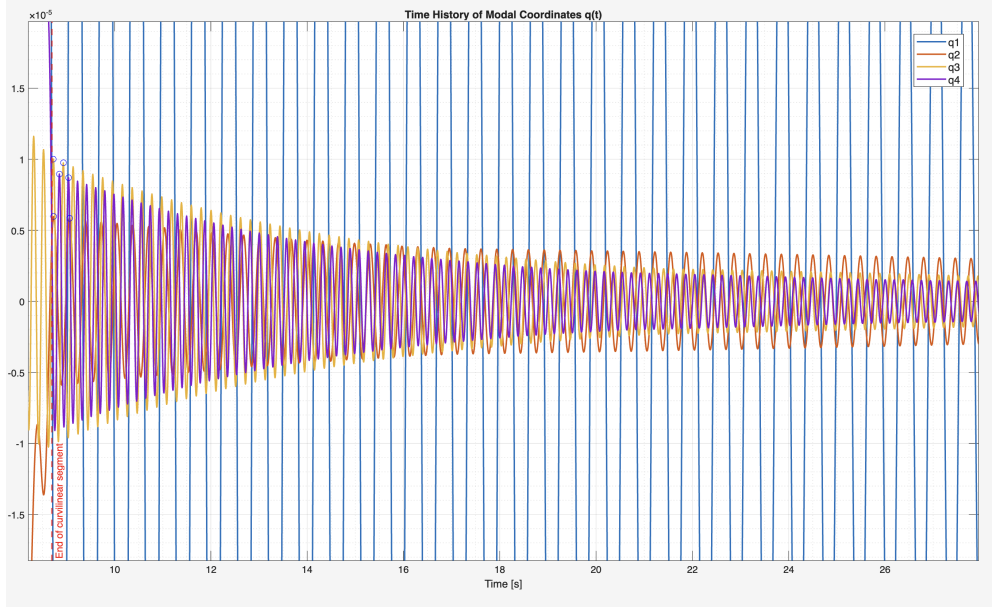


Figure 12

The plots show that the modal coordinate associated with the fourth mode initially exhibits larger oscillation amplitudes than those associated with the second and third modes. This behavior is likely due to the fact that the fourth mode is more strongly excited during the final stage of the skier's motion along the jump. However, as time progresses, the effect of damping becomes increasingly significant: the energy associated with this mode is dissipated more rapidly, leading to a faster decay of its oscillation amplitude. As a result, after some time, the peaks of the fourth modal coordinate become smaller than those of the other modal coordinates.

Finally, the actual vertical displacement of the tip of the ski jump can be visualized as a function of time.

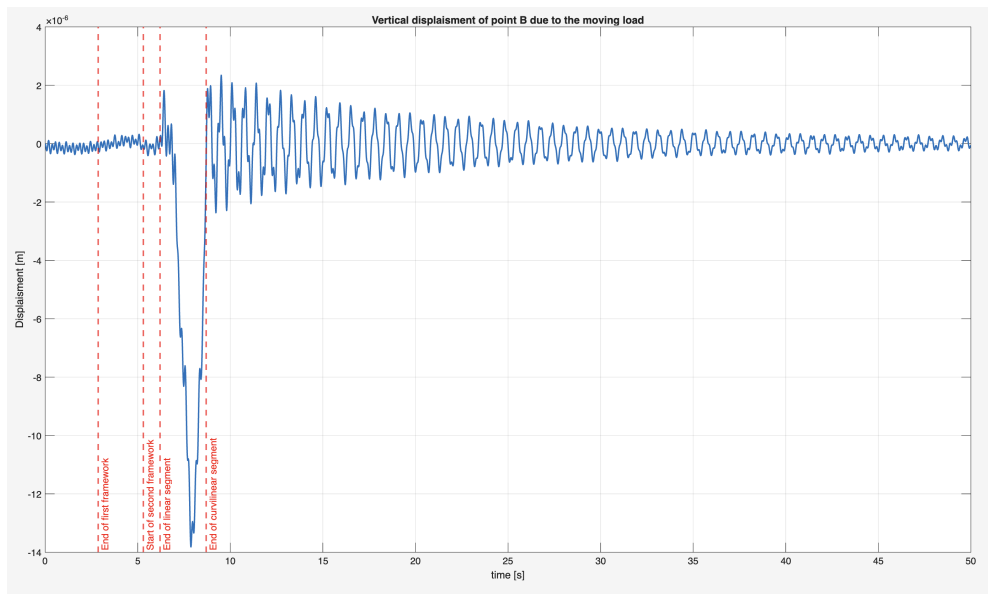


Figure 13

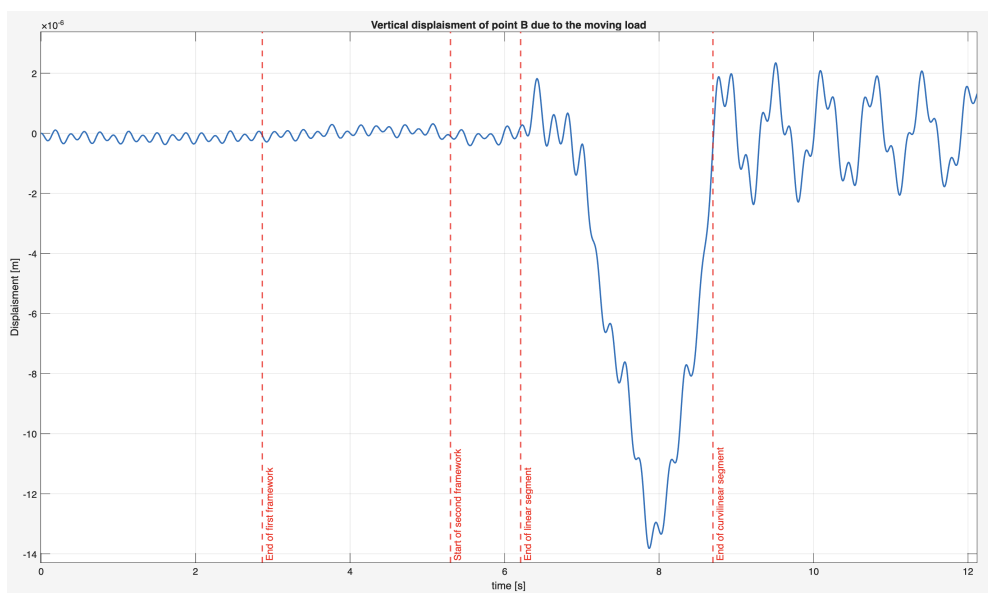


Figure 14